LaRa

Low-Cost Adaptive Robotic-AI Therapy System

Regulated Deterministic AI for Neurodiverse Children

Safety • Structure • Adaptation • Personalization

A privacy-first, fully offline, deterministic AI companion for therapeutic engagement

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01 | Executive Summary

LaRa is a fully offline, safety-first adaptive AI therapy companion engineered specifically for neurodiverse children, with an initial focus on children with Down syndrome. Unlike commercial chatbots or generic educational software, LaRa operates as a regulated, deterministic system in which every behavioral decision — difficulty, pacing, emotional response, and reinforcement — is controlled by transparent, explainable Python logic, not by an unconstrained language model.

The system integrates real-time speech recognition, multi-signal emotion modeling, structured memory, and constrained language generation into a coherent pipeline that runs entirely on local hardware with no cloud dependency and no transmission of sensitive child data. LaRa is designed to be deployed in schools, therapy centers, and homes — anywhere a consistent, patient, and adaptive therapeutic presence is needed but qualified human time is limited or unavailable.

LaRa is not a product built on top of a chatbot. It is a purpose-built therapeutic platform in which AI plays a precisely bounded, safety-governed role.

100%	6	0	5+
Offline / Local Processing	Layered Memory System	Transcripts Ever Stored	Adaptive Subsystems

02 | The Problem We Are Solving

The Gap in Therapeutic Access

Children with Down syndrome and other neurodiverse conditions experience the world differently — their cognitive processing, emotional regulation, attention spans, and communication patterns require patient, consistent, and adaptive interaction. Research consistently demonstrates that frequent, structured one-on-one therapeutic engagement produces measurable improvements in language, numeracy, and social development. However, the infrastructure to provide this at scale does not exist.

> Qualified therapy time is scarce and expensive — typically rationed to hours per week rather than daily engagement.
> Most educational software is static; it cannot adapt to real-time emotional state or cognitive load.
> Mainstream AI assistants are inherently unpredictable, creating serious safety risks for emotionally vulnerable children.
> Cloud-based AI raises significant and unresolved data privacy concerns when children are the subjects.
> The majority of assistive technology does not distinguish between content delivery and therapeutic support.

Why Existing Solutions Are Insufficient

Solution Type	Critical Limitation
Standard educational software	Rigid, non-adaptive, no emotional awareness
General-purpose AI chatbots	Unpredictable outputs — unsafe for neurodiverse children
Cloud-based AI assistants	Transmits sensitive data, privacy risks, latency issues
Human therapy alone	Expensive, scarce, cannot scale to daily interaction
Simple robotics platforms	No language intelligence, cannot generalize or adapt

03 | What LaRa Is

A Regulated Therapeutic AI System — Not a Chatbot

LaRa is engineered around a single governing principle: the LLM (large language model) is a renderer, not a decision-maker. Every critical behavioral decision in the system — when to increase or decrease task difficulty, how to respond to emotional distress, which encouragement style to apply, how fast to speak — is made by deterministic Python logic that can be inspected, tested, and explained. The LLM reads structured guidance and generates safe, appropriately paced language. It never drives behavior independently.

> Core Design Hierarchy:
>
> Safety > Structure > Adaptation > Personalization
>
> Every system component is ordered by this hierarchy. No adaptation, personalization, or feature richness may ever compromise safety or structural predictability.

Primary Design Commitments

- Predictability over randomness — the child always knows what to expect
- Clarity over complexity — one instruction at a time, simple vocabulary
- Stability over responsiveness — gradual, threshold-gated adaptation only
- Gentle pacing over intensity — slower defaults, regulated speech speed
- Low cognitive load over feature richness — minimal stimuli, consistent patterns
- Privacy by design — zero cloud processing, no transcripts, local-only operation

04 | System Architecture

High-Level Pipeline

LaRa processes each child interaction through a structured, sequential pipeline. Every stage has defined inputs, outputs, and safety constraints. No stage may be bypassed or reordered.

Stage	Component	Function
1 — Perception	Whisper STT (local, small.en)	Transcribes 16kHz speech to text; WebRTC VAD detects voice activity
2 — Emotion	MoodDetector	Text + audio prosody analysis; 3-turn consensus filter; outputs mood + confidence score
3 — Session Pre	SessionState.update_pre()	Updates mood streaks, difficulty cooldown, and turn counter before any decision
4 — Regulation	RegulationState	Normalizes mood persistence into a 0.0–1.0 score governing behavioral intensity
5 — Difficulty	DifficultyGate	Evaluates streak thresholds; applies +1 / −1 / lock to task difficulty level
6 — Strategy	RecoveryStrategy Manager	Translates regulation score to TTS speed, response length, and LLM guidance text
7 — Reinforcement	Reinforcement Adaptation Manager	Selects encouragement style from 4 options; learns effectiveness over time
8 — Mode Control	ModeController	Determines Therapy Mode or Exploration Mode for the current turn (sovereign)
9 — Generation	LLM via Ollama (local)	Receives 7-part structured prompt; generates language — renders, never decides
10 — Speech	Kokoro TTS — af_bella voice	Synthesizes audio at strategy-controlled speed; supports barge-in interruption
11 — Session Post	SessionState.update_post()	Increments turn count; triggers learning progress update and memory persistence
12 — Memory	UserMemory (SQLite, local)	Persists aggregated metrics only — never raw transcripts, never emotional labels

Operational Modes

Therapy Mode (Default)

In Therapy Mode, the system exercises full deterministic control. The LLM is restricted to rendering language only — no autonomous tool selection is permitted. Strict pacing and regulation are enforced. Mood signals directly influence strategy selection and difficulty adaptation is active and gated.

Exploration Mode

Exploration Mode is triggered only by an explicit user intent signal. In this mode, the LLM may suggest a single validated tool call. One tool per turn is the hard limit. Recursive agent loops and tool chaining are architecturally prohibited. The ModeController is the sole arbiter of mode — the LLM cannot initiate or sustain transitions independently.

05 | Memory Architecture

Philosophy

LaRa's memory system is structured behavioral infrastructure — not a conversation log. It exists to preserve educational progress and emotional trend metrics across sessions, not to reconstruct dialogue or build personality profiles. Raw transcripts are never stored. Emotional states are aggregated as counts, never labeled permanently.

```
Memory Layers:
  Layer 1   Immediate Turn Buffer   —  ephemeral, cleared per turn
  Layer 2   Rolling Session Buffer  —  last 5 turns, 24-hour expiry
  Layer 3   Session Summary         —  structured, non-narrative, generated per turn
  Layer 4   Structured Session State—  RAM only, expires at session end
  Layer 5   Long-Term User Memory   —  SQLite, aggregated metrics only
  Layer 6   Vector Memory           —  ChromaDB, optional, 90-day expiry
```

Long-Term Memory Schema (SQLite)

Table	Contents
users	User ID, instruction depth baseline, TTS speed preference
learning_progress	Mastery level (0–5) per concept, attempt counts, success rate
emotional_metrics	Frustration / Recovery / Stability event counts with weekly decay (0.95×)
child_preferences	Likes and dislikes (max 20 entries, JSON array, 6-month soft expiry)
reinforcement_metrics	Encouragement style effectiveness — success rate measured per style

No table in this schema stores raw speech, conversation transcripts, or direct emotional labels. Emotional data is stored as event counts only, and these counts are normalized weekly using a 0.95× decay factor to prevent historical events from disproportionately influencing current behavior.

Session Summary Format

Every turn, LaRa generates a structured session summary injected into the LLM prompt. It replaces older raw conversation turns, ensuring the LLM always operates with current context without accumulating stale history:

```
[Session State]
Concept: counting   |   Difficulty: 2/5   |   Turn: 7
```

> Stability trend: improving | Frustration streak: 0 | Stability streak: 3
> Reinforcement: calm_validation | Mastery: 2/5

06 | Adaptive Systems in Detail

Mood Detection

LaRa's emotion layer processes two independent input signals and combines them through a consensus filter to produce a single mood signal with an associated confidence score. This multi-stage approach ensures that transient noise does not trigger behavioral adaptation.

- 6 mood categories: neutral, happy, frustrated, anxious, sad, quiet
- Dual-signal input: text keyword analysis (regex, word-boundary protected) + audio RMS energy
- 3-turn consensus filter: requires agreement across 2 of 3 recent turns before a mood is accepted
- Confidence decay: consecutive neutral turns gradually reduce confidence, returning safely to neutral
- Single-turn inference never triggers behavioral change — temporal smoothing is mandatory

Difficulty Gating

Task difficulty is managed by the DifficultyGate component, which applies strict threshold rules to ensure changes are gradual, stable, and reversible. Oscillation — rapid up/down switching — is architecturally prevented.

Event	Trigger Condition	Action
Difficulty Decrease	2 consecutive frustrated turns + confidence ≥ 0.60	Reduce difficulty by 1 level
Difficulty Increase	3 consecutive stable turns + confidence ≥ 0.60	Increase difficulty by 1 level
Change Cooldown	Immediately after any difficulty change	Lock difficulty for 2 turns minimum
Boundary Enforcement	At all times	Difficulty constrained to range 1–5 (never reaches 0)

Reinforcement Adaptation

LaRa learns which encouragement style is most effective for each individual child over time. The system tracks four distinct approaches and adapts based on measured outcomes — whether the child demonstrates improved engagement or task success after a particular reinforcement style.

- 4 styles: Calm Validation, Praise-Based, Achievement-Based, Playful Encouragement
- Minimum 5 observed events before any style switch is permitted
- 15% improvement threshold required to justify switching

- Maximum one style switch per session — prevents instability
- Default: always begins at Calm Validation (safest, most universally appropriate)
- Preferred style persisted to SQLite only at session end

Child Preference Memory

LaRa builds a lightweight preference profile for each child through natural conversation. Preferences are detected via deterministic regex parsing and woven naturally into LLM responses to increase engagement and emotional comfort.

- Detects expressions from: "I like X", "I hate Y", "My favorite is Z", "I'm scared of X"
- Maximum 20 preferences stored per child — prevents profile bloat
- Deduplication: if a topic is already stored, sentiment is updated rather than duplicated
- Preferences are injected as natural context, not listed mechanically
- 6-month soft expiry on unused preferences

07 | Safety Framework

Non-Negotiable Safety Invariants

The following constraints are architecturally enforced and cannot be overridden by any runtime configuration, operator instruction, or LLM output. They represent the absolute safety floor of the system.

1. The LLM is a renderer, not a decision-maker. Difficulty, mood response, and reinforcement style are determined entirely by deterministic Python code. The LLM generates language only.
2. No emotional labeling. LaRa never expresses direct emotional interpretations to or about the child. Emotional signals are internal processing artifacts only.
3. Gradual adaptation only. Difficulty changes require two or more consecutive threshold-exceeding turns, minimum confidence of 0.60, and a two-turn cooldown after any change.
4. Privacy is absolute. All processing is local. No audio, transcripts, or behavioral data are transmitted to any external service. No biometric data is collected or stored.
5. False negatives preferred. When confidence in any signal is low, the system defaults to neutral supportive behavior rather than acting on uncertain information.
6. Predictability first. One thought at a time, one question at a time, consistent tone throughout every interaction.
7. Task difficulty never reaches zero. A micro-challenge always exists to maintain engagement without overwhelming.
8. Interaction stops when engagement drops critically. The system gracefully pauses rather than persisting with a disengaged child.

Fail-Safe Behavioral Protocol

When the system encounters uncertainty at any level — ambiguous emotion signal, low confidence mood inference, or unexpected input — it follows a defined fail-safe protocol:

> Fail-Safe Protocol (applied whenever confidence is insufficient):
> 1. Slow down the interaction — reduce TTS speed, increase pause duration
> 2. Simplify communication — shorter sentences, smaller vocabulary
> 3. Encourage gently — affirming, non-directive, non-judgmental language
> 4. Never intensify stimulation — no escalation in any dimension
> 5. Maintain safe neutral state until confidence is restored

Ethical Constraints

- The system will never diagnose, make clinical claims, or provide medical advice
- The system will never use emotionally intense, frightening, or negative reinforcement language
- The system will never criticize, demean, or express disappointment toward the child

- All responses must be encouraging, calm, structured, and non-judgmental at all times
- Caregiver oversight is structurally supported through session logging and progress reporting
- No raw conversation data is ever accessible to any party beyond the immediate session

08 | Privacy & Data Architecture

LaRa treats data privacy as a foundational constraint, not a compliance checkbox. Because the primary users are vulnerable children — a population with heightened legal and ethical protection requirements — the system is designed so that sensitive data is never generated, transmitted, or stored in the first place.

Data Category	LaRa's Approach
Raw speech audio	Never stored after transcription — discarded immediately
Conversation transcripts	Not stored anywhere, ever, under any condition
Emotional labels	Never stored as labels — converted to aggregate counts only
Cloud processing	Zero — 100% local inference on-device
Biometric identifiers	Not collected or processed at any point
Session data	Auto-expires and is deleted after 24 hours
Emotional event counts	Normalized weekly with 0.95× decay factor
Story summaries (vector)	Soft-expire after 90 days; max 200 characters each

09 | Vision Perception Module

LaRa has a built-in “set of eyes” that watches the child throughout every session. This vision system runs as a separate program alongside the main LaRa software. Keeping it separate means that even if the camera processing is busy, it never slows down what LaRa is saying or listening to.

DEPLOYMENT ARCHITECTURE

The vision system runs quietly in the background on the same device. The main LaRa program checks in with it regularly — every fraction of a second — to get the latest snapshot of what the camera sees. If the vision system ever needs to restart, the therapy session keeps going uninterrupted.

> **Service** lara-vision-perception | FastAPI | Port 8001
>
> **Isolation** Zero shared memory with main LaRa process. HTTP-only boundary.
>
> **Offline** Fully local. No cloud API calls. Runs on Apple Silicon and ARM hardware.
>
> **Performance Target** 15 FPS stable at 640x480. Full pipeline under 80ms per frame.
>
> **Latency** /latest endpoint: < 10ms response (memory read only, no computation)

Multi-Threaded Pipeline Design

The camera and the analysis work at the same time. One part of the system grabs the latest image from the camera. The other part analyses it. They run side by side so that LaRa is always working with the most up-to-date picture of the child, never a stale one from a few seconds ago. The system is designed to analyse around 15 pictures every second.

A “watchdog” runs silently in the background checking that the vision system is working normally. If the system freezes or gets stuck, the watchdog notices within half a second and automatically restarts the vision pipeline. After a restart it begins fresh, so the system never accidentally reports old information as current.

Frame Quality Gate

Every captured frame passes through a quality gate before entering the detection pipeline. Frames are evaluated for minimum brightness (mean pixel intensity > 30) and minimum sharpness (Laplacian variance > 50). Frames that fail the gate are skipped, but the system does not simply discard the output — it returns the last known stable PerceptionOutput with the skip reason flagged. This ensures the dashboard always receives a meaningful signal rather than silence, even in poor lighting conditions.

If too many images in a row are unusable (more than 10 in a row), LaRa assumes the child is no longer visible and marks them as absent. Engagement scores drop accordingly. This stops the system from mistakenly showing high engagement when the camera simply cannot see anything.

10 | Three Perception Channels

The vision pipeline is organised around three independent detection subsystems. Each operates on every frame (or on a throttled interval for YOLO) and contributes signals to the unified PerceptionOutput. Below is a detailed breakdown of each channel.

Channel 1 — Face Detection & Head Pose Estimation

LaRa uses a face-tracking library to find the child's face in each image. It maps out key points on the face — the nose, chin, eye corners, and mouth corners. Using these six fixed reference points, it can figure out which direction the child's head is pointing, even from a flat camera image.

The system compares those face points to a built-in 3D model of a human head. This calculation produces two angles: left-right (yaw) and up-down (pitch). If both angles are small — meaning the head is roughly pointing at the screen — LaRa records that the child is looking at the screen.

Three-Signal Pose Confidence

Just knowing the head angle is not enough on its own. A child might be partly hidden, turned to the side, or in poor light — all of which can make the angle look normal when the child is actually looking away. So before trusting any gaze reading, LaRa scores its own confidence using three independent checks.

> **Signal 1 — Landmark Visibility (weight: 0.40)** How clearly all six face reference points are visible. If any are hidden or blurry, this score falls and gaze classification is skipped.
>
> **Signal 2 — Reprojection Error (weight: 0.40)** After calculating the head angle, LaRa checks its own work by predicting where the face points should appear and comparing that to where they actually are. A small gap means a trustworthy estimate.
>
> **Signal 3 — Landmark Spread Quality (weight: 0.20)** A check that the six face reference points are spread out naturally across the face. If they are all clustered in one spot, the angle estimate is unreliable and confidence is reduced.

If all three checks together score below 0.40, LaRa plays it safe and records the child as not looking at the screen, even if the raw angle suggests otherwise. The head angle itself is averaged over the last 5 frames to prevent flickering.

Channel 2 — Hand Gesture Recognition

LaRa can also recognise simple hand gestures. It tracks 21 points on the hand and uses straightforward rules to decide which fingers are extended. This is fast and lightweight — it runs on every single frame without slowing the system down.

A finger is counted as extended when its tip is higher than its base knuckle. The thumb is handled slightly differently because it points sideways — and the rule correctly flips for left hands versus right hands, so both work equally well.

The classifier recognises six gesture labels:

- NONE — No hand detected or insufficient confidence
- OPEN_PALM — All five fingers extended

- FIST — No fingers extended
- THUMBS_UP — Thumb extended, all other fingers closed
- POINTING — Index finger extended, others closed
- PEACE — Index and middle fingers extended, others closed

Channel 3 — Environmental Object Detection (YOLO)

LaRa can recognise everyday objects in the room — things like a book, a cup, or a toy. It uses a compact AI model called YOLO to do this. The result is a simple list of object names seen in the current camera view, which helps LaRa understand the child's physical environment.

Object detection is more demanding than face or gesture detection, so it does not need to run on every single frame. LaRa automatically decides how often to run it based on how much spare computing power is available — at least once per frame, at most once every six frames. Between checks it reuses the most recent result.

11 | Adaptive Throttle & Engagement Scoring

YOLO Adaptive Throttle Algorithm

There is a trade-off: checking objects every single frame gives the most accurate results, but slows everything down. Checking too rarely means the information gets stale. LaRa automatically finds the right balance while the session is running — no manual tuning needed.

After each frame, LaRa checks how long everything took. If it took too long, object detection is asked to skip more frames to free up time. If there is headroom, it runs more often. Changes only happen at most every 2 seconds to keep behaviour stable. In between object-detection frames, the last known result is reused.

> **Minimum interval** 1 frame (YOLO runs every frame — maximum accuracy)
>
> **Default interval** 3 frames (YOLO runs every third frame — balanced)
>
> **Maximum interval** 6 frames (YOLO runs every sixth frame — maximum CPU relief)
>
> **Hysteresis band** ±5ms around frame budget (prevents oscillation)
>
> **Adjustment cooldown** 2.0 seconds minimum between interval changes

Dual-Track Engagement Scoring

LaRa calculates engagement as two separate numbers that move at different speeds. The fast score is used internally for decisions — it reacts in under a second. The slow score is shown to caregivers on the dashboard — it changes gradually so the display stays calm and easy to read, not jumping around every second.

Engagement Signal Composition

Each frame, LaRa builds an engagement reading from three simple questions:

> **Face Presence (weight: 0.45)** Is a face visible? This is the most important signal. If the child cannot be seen, the engagement score starts to drop.
>
> **Gaze Confirmation (weight: 0.45)** Is the child facing the screen? Only counted when LaRa is confident the head angle reading is reliable.
>
> **Gesture Activity (weight: 0.10, gaze-gated)** Is the child making a gesture with their hand? Counts fully when they are also looking at the screen. Still contributes a small amount if they are gesturing while looking away — because a child actively moving their hand is likely still engaged.

Each frame's reading is added to a running average of the last 15 frames (about one second of data). This smooths out brief blips so the engagement score reflects a genuine trend rather than a momentary flicker.

Absence Decay

When the child cannot be seen, both scores fall gradually rather than dropping to zero all at once. The decision engine sees the fast drop and can react quickly. The caregiver dashboard shows the gentle fade — a slow, calm decline rather than a sudden alarm.

Example decay curves at 15 FPS after face disappears:

> **After 0.5s (8 frames) — Internal** 0.70^8 = 0.057 (near zero)
>
> **After 0.5s (8 frames) — UI** 0.95^8 = 0.663 (still displaying moderate engagement)
>
> **After 1.0s (15 frames) — UI** 0.95^15 = 0.463 (halved — visible decline, not alarm)

12 | Attention State Machine & Temporal Smoothing

Brief glances away are perfectly normal. A child reaching for a pencil or turning toward a sound should not be logged as distracted. LaRa only records a real attention change when the same behaviour persists for a short but meaningful amount of time — filtering out the noise of everyday movement.

Four Attention States

FOCUSED Child is present and consistently looking at the screen. Requires sustained gaze for a minimum of 5 consecutive frames (~0.33s at 15 FPS) before transitioning into this state.

DISTRACTED Child is present but has looked away from the screen. Requires 8 consecutive frames (~0.53s) of non-gaze before transitioning — brief glances away do not trigger this state.

ABSENT No face detected. The system treats this as a full absence. Engagement scores decay, and distraction frame counters are reset.

UNKNOWN Insufficient frames have been processed since startup or reinitialisation to make a reliable determination.

State Transition Rules

When LaRa notices the child may have looked away, it does not immediately log this. Instead it waits and counts how many frames in a row confirm the same thing. Only once enough frames confirm the new behaviour does LaRa officially record the change. This is called a confirmation window, and it protects against false alarms.

This is especially important for children with Down syndrome, who may naturally look away more often as they think or process information. LaRa is designed to treat these brief diversions as normal — not as problems. Only sustained, prolonged inattention is recorded as distraction.

Distraction Frame Counting

LaRa also counts how many frames the child has been distracted in a row. Dividing this by the frame rate gives the exact number of seconds. This counter resets the moment the child looks back at the screen, so each new distraction episode is measured independently.

Memory Management & Watchdog

A therapy session can last 30 minutes or more. During that time LaRa keeps a close eye on how much computer memory it is using. It checks this every 2 seconds (every 30 frames). If memory starts creeping up in a worrying pattern, it logs a warning so the issue can be investigated before it affects a real session.

LaRa tracks memory trends over the last 20 measurements. A steady rise that continues for three windows in a row triggers a warning. It also watches for whether each new session uses more

memory than the last. Both patterns are early signs of trouble, flagged automatically so developers can fix them before any child is affected.

System Confidence Score

Every PerceptionOutput includes a systemConfidence score (0.0 to 1.0) representing the overall reliability of the current perception snapshot as a weighted fusion of the three per-channel confidence values:

> **Face Confidence (weight: 0.40)** Scores how well-positioned the face is in the frame. A child looking straight ahead in the centre scores highest.
>
> **Pose Confidence (weight: 0.35)** Combines all three face-quality checks into one score. Higher means LaRa is more confident about where the child is looking.
>
> **Object Confidence (weight: 0.25)** Average confidence score for all objects detected. If no objects are visible, this is left out of the overall score entirely.

The system confidence is logged with every session record and exposed in the live monitoring dashboard. Values below 0.5 trigger a visual reliability warning to alert caregivers that perception quality may be degraded.

13 | PerceptionOutput Schema & API

Every frame produces one structured snapshot of what LaRa can see. This snapshot — called a PerceptionOutput — is a fixed list of fields that never changes unexpectedly. It is shared with the decision engine, the session recorder, the caregiver dashboard, and the child's screen. Every part of the system reads from this same single source of truth.

Complete Field Reference

PRESENCE & GAZE FIELDS

presence (bool) True if a human face is detected in the current frame.

faceVerified (bool) True if LaRa has confirmed this face matches the enrolled child. Checked once per second. False if no child profile has been set up for this device.

lookingAtScreen (bool) True if the child's head is facing the screen and the pose reading is reliable enough to trust (combined confidence score of at least 0.40).

attentionState (str) The child's current attention label — FOCUSED, DISTRACTED, ABSENT, or UNKNOWN — after filtering out brief moment-to-moment changes.

distractionFrames (int) How many frames in a row the child has been distracted. Divide by 15 to get seconds. Resets to zero when the child looks back at the screen.

ENGAGEMENT FIELDS

engagementScore (float, 0–1) The engagement level used internally for decisions. Reacts fast — drops noticeably in under half a second if the child looks away.

engagementScoreUI (float, 0–1) The engagement number shown to caregivers on the dashboard. Changes gradually so the display stays calm and readable, not jumping around every second.

GESTURE & OBJECT FIELDS

gesture (str) What hand gesture LaRa can see: NONE, OPEN_PALM, FIST, THUMBS_UP, POINTING, or PEACE. Works correctly for both left and right hands.

detectedObjects (tuple[str]) A list of objects visible in the room — for example ["book", "cup"]. Reuses the most recent scan if object detection has not run on this exact frame.

CONFIDENCE FIELDS

confidence.face (float) How centred the face is in the camera frame. A face directly in front scores higher than one at the edge.

confidence.pose (float) Combined score for how trustworthy the head angle estimate is, based on face visibility, calculation accuracy, and landmark spread.

confidence.gesture (float) How confident LaRa is that it correctly identified the hand gesture. 0.0 means no hand was seen.

confidence.objects (float) Average confidence across all detected objects. 0.0 when no objects are visible.

systemConfidence (float) A single overall reliability score for this frame — combining face quality, head pose accuracy, and object detection confidence. Shows how much to trust the full snapshot.

SKIP & METADATA FIELDS

skipped.quality (bool) True if the camera image was too dark or too blurry to use. The previous reliable result is shown instead.

skipped.throttle (bool) True if object detection did not run this frame to save CPU time. The most recent object list is reused.

skipped.camera_drop (bool) True if the camera failed to deliver a frame at all. Rare — usually recovers on the next frame.

timestamp (float) The exact time this snapshot was created, accurate to 0.1 milliseconds.

API Endpoints

The microservice exposes five HTTP endpoints:

GET /health Liveness probe. Returns {status: ok, service: lara-vision-perception}. No engine interaction.

GET /status Engine diagnostics: state (STOPPED/RUNNING/ERROR), FPS, memory_mb, memory_growth_rate_mb_per_sec, memory_leak_suspected, stall_count, yolo_interval. Does not return perception data.

POST /start Opens the camera and starts the pipeline thread. Idempotent — returns already_running if engine is active.

POST /stop Cleanly shuts down the pipeline, joins all threads, and releases the camera device. Idempotent.

GET /latest Returns the most recent PerceptionOutput as JSON. Response time < 10ms (memory read only — no inference). Returns HTTP 503 if engine is not running.

Sample /latest Response


```
  "presence": true,
  "faceVerified": false,
  "lookingAtScreen": true,
  "engagementScore": 0.8720,
  "engagementScoreUI": 0.9140,
  "attentionState": "FOCUSED",
  "distractionFrames": 0,
  "gesture": "THUMBS_UP",
  "detectedObjects": ["book", "cup"],
  "systemConfidence": 0.847,
  "confidence": { "face": 0.912, "pose": 0.783, "gesture": 0.961, "objects": 0.741
},
  "skipped": { "quality": false, "throttle": false, "camera_drop": false },
  "timestamp": 1741022400.3124
```

14 | Technology Stack

Component Overview

Component	Technology	Rationale
Speech-to-Text	OpenAI Whisper small.en (local)	Accurate, fast, fully offline, no cloud dependency
Language Model	Ollama — AgentricAI_TLM 2B	128K context, local inference, low hardware requirements
Text-to-Speech	Kokoro-82M — af_bella voice	Natural prosody, controllable speed, lightweight, local
Long-Term Memory	SQLite (local file)	No server required, simple, reliable, fully inspectable
Vector Memory	ChromaDB (local embeddings)	Semantic story recall without any cloud dependency
Voice Activity	WebRTC VAD	Battery-efficient, precise, low latency
Language	Python 3.11+	Compatibility with all core scientific and ML libraries
Hardware Target	macOS / Low-cost ARM hardware	Designed for accessible, real-world deployment environments

Core System Files

File	Role
whispercpp_STT.py	Main orchestrator — microphone pipeline, VAD, wake word, turn control
mood_detector.py	Text and audio prosody mood analysis with 3-turn consensus filter
session_state.py	RAM-based session with pre/post update phases
regulation_state.py	Normalizes mood streaks into 0.0–1.0 persistence score
recovery_strategy.py	Translates mood regulation into TTS speed, response length, guidance text
reinforcement_manager.py	Adapts encouragement style based on historical effectiveness data

File	Role
session_summary.py	Generates structured, non-narrative session context for LLM prompt
child_preferences.py	Detects and stores child preferences; injects into LLM context
vector_memory.py	ChromaDB RAG for story and theme continuity across sessions
user_memory.py	SQLite persistence layer — aggregated metrics, mastery, preferences
AgentricAi/AgentricTLM.py	LLM interface — strict 7-part prompt builder, streaming response
AgentricAi/kokoro_TTS.py	Local Kokoro TTS with speed control and barge-in interruption support

15 | LLM Interaction Design

The 7-Part Structured Prompt

Every turn, before the LLM generates a single word, it receives a fully structured 7-part prompt assembled by deterministic system logic. This prompt encodes all relevant behavioral guidance, session state, memory context, and interaction history. The LLM's role is to transform this structured guidance into appropriately paced, gentle, child-safe language.

Prompt Part	Content
Part 1 — System Rules	8 core behavioral constraints: identity, tone, and absolute prohibitions
Part 2 — Strategy Guidance	Recovery strategy text from RegulationState (e.g., 'Break into tiny single steps')
Part 3 — Reinforcement Style	Active encouragement style and associated language guidance patterns
Part 4 — Child Context	Known preferences and past story themes woven as natural context
Part 5 — Session Summary	Structured session state: concept, difficulty, turn count, streaks, mastery
Part 6 — Conversation History	Last 5 turns from the rolling session buffer (150 chars per turn, truncated)
Part 7 — Current Message	The child's current utterance as transcribed by Whisper STT

Sample Prompt Construction (Annotated)

```
You are LaRa — a gentle, highly predictable, encouraging therapy assistant for children with Down syndrome.
[8 behavioral constraints: simple sentences, one instruction at a time, no negative reinforcement...]

[Behavioral Guidance]: The child seems to be finding things difficult.
  Break instructions into tiny, single steps. Slow down your response.

[Reinforcement Style]: Use calm, grounding validation phrases.

[Child Preferences]: Likes: dinosaurs, blue, blocks.  Dislikes: loud sounds.

[Session State]
  Concept: counting   |   Difficulty: 2/5   |   Turn: 4
  Stability trend: declining   |   Frustration streak: 2   |   Stability streak: 0
  Reinforcement: calm_validation   |   Mastery: 1/5

--- Recent Conversation ---
```

```
  User:  Can we learn counting?
  LaRa:  Of course! Let us count together. Can you say one?
  User:  One!
  LaRa:  Wonderful! That was perfect. Now can you say two?
  User:  This is too hard.
  LaRa:  That is okay. We can go slower. Let us just do one at a time.
--- End of History ---

User says: I still can't do it.
LaRa says:
```

16 | Implementation Roadmap

LaRa is developed across six structured phases, each building on the stability and correctness of the previous. This phased approach ensures safety-critical components are validated before dependent systems are introduced.

Phase	Focus Area	Key Deliverables
Phase 1 — Stability Core	Foundation architecture	SessionState pre/post separation; RegulationState streak persistence; DifficultyGate isolation
Phase 2 — Memory Enhancement	Structured memory system	Token-based rolling buffer; deterministic SessionSummarizer; structured LLM context injection
Phase 3 — Hybrid Mode System	Controlled tool use	ModeController; ToolRegistry; ToolExecutionGuard; structured function-calling protocol
Phase 4 — Reinforcement Adapt.	Personalized encouragement	Style tracking metrics; minimum-event threshold enforcement; per-session lock
Phase 5 — Safety Hardening	Constraint enforcement	Stimulation caps for media tools; difficulty oscillation cooldown; tool invocation rate limiting
Phase 6 — Dashboard Integration	Caregiver visibility (optional)	Session summary export; mastery trend visualization; emotional stability reporting (no labels)

17 | Use Cases & Deployment Contexts

Primary Deployment Environments

- Special education classrooms — daily structured learning companion for neurodiverse students
- Therapeutic centers — between-session engagement tool supervised by therapists
- Home environments — consistent after-school companion supporting learned skills
- Pediatric hospital settings — low-stimulation, patient-paced engagement for hospitalized children

Interaction Scenarios

Structured Learning Session

A therapist or educator initiates a session around a target concept such as counting, colors, or shapes. LaRa guides the child through progressively structured exercises, automatically adjusting difficulty in response to real-time emotional and performance signals. The session is self-pacing and self-regulating. At session end, aggregated metrics are persisted for caregiver review.

Distress Recovery

If the child expresses frustration, LaRa's RegulationState detects the persistence of the signal across multiple turns and triggers a recovery strategy — slowing TTS speed, simplifying language, reducing task demands, and shifting to purely supportive language. The system never escalates during a distress event.

Free Exploration

When the child expresses curiosity about a topic outside the current structured task, Exploration Mode may activate. LaRa may suggest a validated media retrieval within strict execution constraints — one tool per turn, no chaining, no autonomous escalation.

Scalability

LaRa's architecture is user-isolated by design. Every child is identified by a unique user_id, and all memory layers — SQLite rows, vector collections, session state — are fully partitioned. A single LaRa device can serve multiple children with completely independent behavioral profiles, preference memories, and learning progress records. No child's data influences another child's system behavior.

18 | Future Development

Near-Term Priorities

- Therapist Dashboard: Export session summaries as caregiver-facing reports showing mastery trends, emotional stability metrics, and concept progress — no transcripts, no raw emotional labels
- Curriculum Integration: Connect LearningProgressManager to a structured concept curriculum (colours → numbers → shapes → words → sentences)
- Multi-User Device Support: Formalize the multi-child architecture with device-level user management and session handover protocols
- Hardware Portability: Validate deployment on low-cost ARM hardware (Raspberry Pi 5, Apple Silicon) for accessible real-world installation

Medium-Term Vision

- Physical Robot Integration: Mount the LaRa software stack on a low-cost social robot platform, adding facial expression and gesture responses governed by the same deterministic safety framework
- Multilingual Support: Extend Whisper and Kokoro integrations to support additional languages for non-English speaking neurodiverse populations
- Formal Clinical Evaluation: Partnership with educational psychology or pediatric therapy research groups to conduct structured efficacy studies
- Accessibility Certification: Pursue formal review against relevant assistive technology and child safety standards

19 | Closing Statement

LaRa exists because neurodiverse children deserve access to consistent, patient, and adaptive therapeutic support — not as an occasional resource, but as a daily companion that understands the importance of emotional safety, predictability, and gentle progression.

Every design decision in LaRa reflects a deliberate rejection of the dominant trend in AI development — the trend toward autonomy, surprise, and open-ended capability. LaRa is deliberately constrained, deliberately transparent, and deliberately conservative. Its LLM is subordinate to its safety logic. Its memory stores measurements, not interpretations. Its adaptive systems move slowly and require evidence before acting.

This is not a limitation of ambition. It is the expression of a fundamental conviction: that in therapeutic contexts involving vulnerable children, predictability is a feature, not a constraint. Caution is intelligence. Safety is the product.

LaRa is structured behavioral intelligence.

It enables growth, stability, and continuity without sacrificing safety.

Safety • Structure • Adaptation • Personalization

LaRa | Low-Cost Adaptive Robotic-AI Therapy System
Official Project Document — Version 2.0 | 2025